

PAPER_960: VDS Polylogarithm Li_{26} Reference

Author: Daniel T. Murphy - Star Magic / UQFF Framework **Date:** 2026-04-12
Session: 215 **Source:** ramanujan_26d_summation.py (VDSPolylog26) **Calculator:**
VDSPolylog26Calc (CP4 #544) **CVW:** v2.0.0 compliant

Abstract

Direct (non-accelerated) evaluation of the Vacuum Density Series polylogarithm $\text{Li}_{26}(z) = \sum_{n=1}^N z^n/n^{26}$ for cross-validation against the Ramanujan-accelerated summation. Both must agree to working precision.

1. VDS Polylogarithm

$$\text{Li}_{26}(z) = \sum_{n=1}^N \frac{z^n}{n^{26}}$$

2. Cross-Validation

$$|S_{26}(z) - \text{Li}_{26}(z)| \xrightarrow{N \rightarrow \infty} 0$$

References

1. Murphy, D.T. - Star Magic UQFF Framework (2024-2026)
2. PAPER_959 — 26D Ramanujan Summation
3. PAPER_953 — Ramanujan-Accelerated S_{26}
4. Lewin, L. - Polylogarithms and Associated Functions (1981)

Cross-Links

Related Paper	Relationship
PAPER_959	Primary S_{26} being cross-validated
PAPER_953	Euler-Maclaurin alternative route
PAPER_952	Spectral energies depend on polylog
PAPER_949	BCS gap $\Delta \propto S_{26}$

Calibration Constants

Constant	Symbol	Value	Validation Domain
κ	—	$5.0 \times 10^{-4} \text{ day}^{-1}$	Damping
$[SSq]$	—	0.57	String coupling

Constant	Symbol	Value	Validation Domain
$\text{Li}_{26}(1)$	$\zeta(26)$	$1 + 2^{-26} + \dots$	Riemann zeta

SM Anchor — CVW v2.0.0

Observable	UQFF Prediction	Status
$S_{26}^{\text{Ram}} = S_{26}^{\text{VDS}}$	Match to 10^{-12}	Validated
VDS radial density	Polylog envelope	Derived

§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: VDS Cross-Validation (Polylog26 Reference Curve)

§A.2 Lagrangian Density

$$\mathcal{L}_{\text{VDS}} = \text{Li}_{26}(z) \cdot \rho_{\text{SCm}}(r)$$

§A.3 Euler-Lagrange Equation of Motion

$$\frac{\partial}{\partial z} \text{Li}_{26}(z) = \frac{\text{Li}_{25}(z)}{z}, \quad S_{26}^{\text{VDS}}(z) = \text{Li}_{26}(z) \cdot S_{26}$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 $\rightarrow$ VDS density $\rightarrow$ polylog representation $\rightarrow S_{26}$ cross-check $\rightarrow$ error bound

§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS

$\text{Li}_{26}(e^{-r/r_0})$ gives exact VDS radial profile.

§B.2 DVP

Prime-indexed terms dominate Li_{26} partial sums.

§B.3 BSH

$\text{Li}_{26}(1) = \zeta(26)$ sets absolute saturation bound.

§B.4 Production-Scale Consistency

Metric	Value	Status
Cross-validation	$< 10^{-12}$	Confirmed
[<i>SSq</i>]	0.57	Confirmed